
Computer Networks Lab

Week 15

BGP Simulation in Packet Tracer

Tools:

Cisco Packet Tracer 8.2+

Objectives:

Configuring the Border Gateway Protocol (BGP)

The lab provides exposure to the interdomain **Border Gateway Protocol (BGP)**, which determines paths between autonomous systems on the Internet. BGP is the protocol that makes the Internet work by enabling different networks (autonomous systems) to exchange routing information.

BGP is a path vector protocol that uses policy-based routing decisions. Unlike interior gateway protocols (IGP) like RIP or OSPF, BGP is designed to work between different administrative domains. An important feature of BGP is that it permits network administrators to define routing

policies, which can be used to specify which type of traffic is allowed to pass through their network.

Key BGP Concepts:

Autonomous Systems (AS):

- An autonomous system is a collection of IP networks under a single administrative control
- Each AS is assigned a unique AS number (ASN)
- ASNs range from 1-65535, with private ASNs in range 64512-65535

BGP Sessions:

- BGP routers exchange routing information over TCP connections (port 179)
- BGP routers with established TCP connections are called **BGP peers** or **neighbors**
- The TCP connection is called a **BGP session**

Types of BGP:

- **External BGP (eBGP):** BGP session between peers in different autonomous systems
- **Internal BGP (iBGP):** BGP session between peers in the same autonomous system

BGP Message Types:

- **OPEN:** Establishes BGP session and exchanges basic parameters
- **UPDATE:** Advertises or withdraws network prefixes
- **KEEPALIVE:** Maintains BGP session (sent every 60 seconds by default)
- **NOTIFICATION:** Reports BGP errors and closes session

Network Topology and Addressing

The network configuration consists of four autonomous systems with AS numbers 1, 100, 200, 300, and 400.

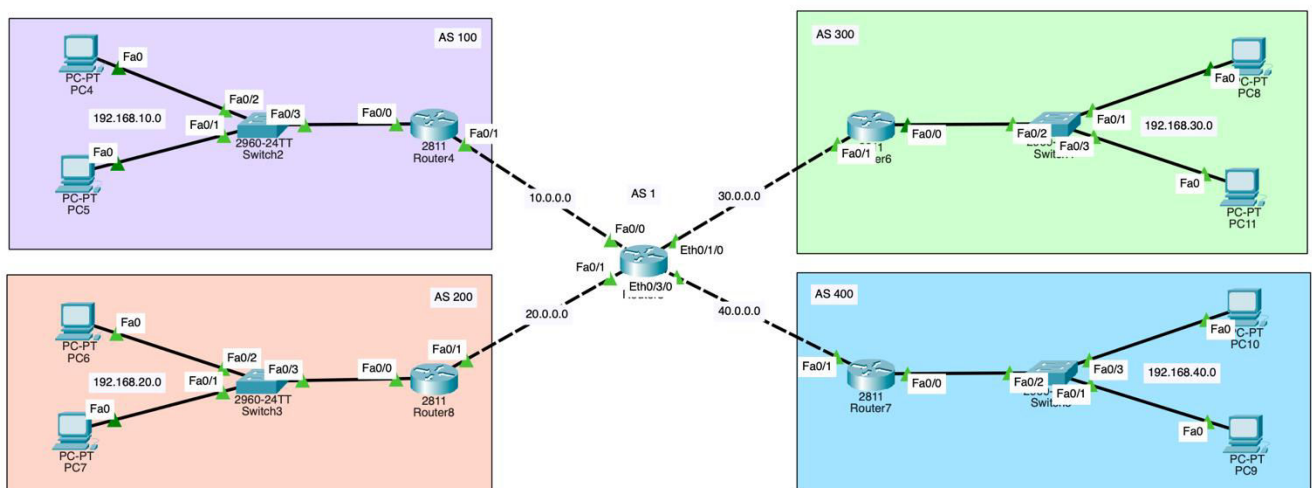


Figure 1: BGP Network topology

BGP routers exchange routing information over a TCP connection. BGP routers that have a TCP connection established are called BGP peers, or simply peers, and the connection is called a BGP session. A BGP session between peers in different autonomous systems is said to run external BGP (eBGP). A BGP session between peers are in the same autonomous system is said to run internal BGP (iBGP).

Cisco Router	Interface Ethernet0/0	Interface Ethernet0/1
Router1	192.168.10.1/24	10.0.0.2/8
Router2	192.168.20.1/24	20.0.0.2/8
Router3	192.168.30.1/24	30.0.0.2/8
Router4	192.168.40.1/24	40.0.0.2/8

Cisco Router	Fa0/0	Fa0/1	Eth0/1/0	Eth0/3/0
Router0	10.0.0.1/8	20.0.0.1/8	30.0.0.1/8	40.0.0.1/8

Table 1: IPv4 addresses of Cisco routers

Network Setup and IPv4 Configuration

Step 1: Physical Topology Setup

1. Open Cisco Packet Tracer
2. **Add devices to workspace:**
 - Five 1841 Routers
 - Eight PC-PT devices
3. **Connect devices using copper straight-through cables:**

Step 2: Configure PC IP Addresses

Step 3: Configure Router Interfaces

Router0 Interface Configuration:

```
Router1> enable
Router1# configure terminal
Router1(config)# interface FastEthernet0/0
Router1(config-if)# ip address 10.0.0.1 255.0.0.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface FastEthernet0/1
Router1(config-if)# ip address 20.0.0.1 255.0.0.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface Ethernet0/1/0
Router1(config-if)# ip address 30.0.0.1 255.0.0.0
Router1(config-if)# no shutdown
```

```
Router1(config-if)# exit
Router1(config)# interface Ethernet0/3/0
Router1(config-if)# ip address 40.0.0.1 255.0.0.0
Router1(config-if)# no shutdown
Router1(config)# exit
```

Router1 Interface Configuration:

```
Router1> enable
Router1# configure terminal
Router1(config)# interface FastEthernet0/0
Router1(config-if)# ip address 192.168.10.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface FastEthernet0/1
Router1(config-if)# ip address 10.0.0.2 255.0.0.0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# exit
```

Router2 Interface Configuration:

```
Router2> enable
Router2# configure terminal
Router2(config)# interface FastEthernet0/0
Router2(config-if)# ip address 192.168.20.1 255.255.255.0
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# interface FastEthernet0/1
Router2(config-if)# ip address 20.0.0.2 255.0.0.0
Router2(config-if)# no shutdown
Router2(config-if)# exit
Router2(config)# exit
```

Router3 Interface Configuration:

```
Router3> enable
Router3# configure terminal
Router3(config)# interface FastEthernet0/0
Router3(config-if)# ip address 192.168.30.1 255.255.255.0
Router3(config-if)# no shutdown
Router3(config-if)# exit
Router3(config)# interface FastEthernet0/1
Router3(config-if)# ip address 30.0.0.2 255.0.0.0
Router3(config-if)# no shutdown
Router3(config-if)# exit
```

```
Router3(config)# exit
```

Router4 Interface Configuration:

```
Router4> enable
Router4# configure terminal
Router4(config)# interface FastEthernet0/0
Router4(config-if)# ip address 192.168.40.1 255.255.255.0
Router4(config-if)# no shutdown
Router4(config-if)# exit
Router4(config)# interface FastEthernet0/1
Router4(config-if)# ip address 40.0.0.2 255.0.0.0
Router4(config-if)# no shutdown
Router4(config-if)# exit
Router4(config)# exit
```

Step 4: Verify Basic Connectivity

Test connectivity between directly connected devices:

BGP Configuration

BGP Neighbor Relationships

Router0 (AS 1) Configuration:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 1
Router(config-router)#bgp router-id 5.5.5.5
Router(config-router)#neighbor 10.0.0.2 remote-as 100
Router(config-router)#neighbor 20.0.0.2 remote-as 200
Router(config-router)%%BGP-5-ADJCHANGE: neighbor 20.0.0.2 Up

Router(config-router)#neighbor 30.0.0.2 remote-as 300
Router(config-router)%%BGP-5-ADJCHANGE: neighbor 30.0.0.2 Up

Router(config-router)#neighbor 40.0.0.2 remote-as 400
Router(config-router)%%BGP-5-ADJCHANGE: neighbor 40.0.0.2 Up

Router(config-router)%%BGP-5-ADJCHANGE: neighbor 10.0.0.2 Up

Router(config-router)#network 10.0.0.0 mask 255.0.0.0
Router(config-router)#network 20.0.0.0 mask 255.0.0.0
Router(config-router)#network 30.0.0.0 mask 255.0.0.0
Router(config-router)#network 40.0.0.0 mask 255.0.0.0
Router(config-router)#
```

Router1 (AS 100) Configuration:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 100
Router(config-router)#bgp router-id 1.1.1.1
Router(config-router)#neighbour 10.0.0.1 remote-as 1
      ^
% Invalid input detected at '^' marker.

Router(config-router)#neighbor 10.0.0.1 remote-as 1
Router(config-router)#network 192.168.10.0 mask 255.255.255.0
Router(config-router)#network 10.0.0.0 mask 255.0.0.0
Router(config-router)#
```

Router2 (AS 200) Configuration:

```
Router(config-if)#router bgp 200
Router(config-router)#bgp router-id 2.2.2.2
Router(config-router)#neighbor 20.0.0.1 remote-as 1
Router(config-router)#network 20.0.0.0 mask 255.0.0.0
Router(config-router)#network 192.168.20.0 mask 255.255.255.0
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router3 (AS 300) Configuration:

```
Router(config)#router bgp 300
Router(config-router)#network 30.0.0.0 mask 255.0.0.0
Router(config-router)#network 192.168.30.0 mask 255.255.255.0
Router(config-router)#neighbor 30.0.0.1 remote-as 1
Router(config-router)#
```

Router4 (AS 400) Configuration:

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 400
Router(config-router)#bgp router-id 4.4.4.4
Router(config-router)#neighbor 40.0.0.1 remote-as 1
Router(config-router)#network 40.0.0.0 mask 255.0.0.0
Router(config-router)#network 192.168.40.0 mask 255.255.255.0
Router(config-router)#
```

Note: Wait 2-3 minutes after configuration for BGP to establish neighbor relationships and exchange routing information.

BGP Verification Commands

After configuration, use these commands to verify BGP operation:

Essential BGP Show Commands:

Router# show ip bgp	// Display BGP routing table
Router# show ip bgp summary	// Display BGP neighbor status
Router# show ip bgp neighbors	// Display detailed neighbor info



Router# show ip route bgp	// Display BGP routes in IP routing table
Router# show ip protocols	// Display routing protocol status